

PharmaFlowAI

Project Proposal for UCS503P

Submitted to: Miss. Anushka Chaudhary

Thapar Institute of Engineering and Technology

TEAM MEMBERS:

Sidak Raj Viridi 1024240043

Takshit Mahajan 1024240056

Sahil Goyal 1024240063

Ridhmann Arora 1024240059

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1 Higher-order goal

PharmaFlowAI is planned as a pharmacy retail management system for a small independent pharmacy store. The long-term idea is to provide a single system for managing medicines, stock, expiry, billing, suppliers, customer retail records and day-to-day store operations. The system will also provide sales analytics and later support features such as machine learning and a chatbot to help the owner understand and manage the store more efficiently.

The immediate engineering goal is to build the core pharmacy workflow first. The system will maintain medicines, suppliers, batches, customers, bills and stock movements in a structured database. The design will be kept simple at first and improved after the basic workflow is working.

2 Time-to-value

- We will deliver meaningful increments quickly by prototyping the core pharmacy workflow, starting with medicine, supplier, batch and inventory management and validating it through practical store scenarios.
- We will prioritize fast feedback over perfection by building the system in short iterations, gradually integrating billing, customer records and sales tracking as the core workflow becomes stable.
- Success is defined by what can be delivered and measured in each iteration, starting with a working inventory workflow and then improving the system through testing and feedback before adding analytics, ML, and chatbot features.

3 Problem Statement

A small pharmacy needs to manage medicines, inventory, suppliers, billing and customer transactions while keeping track of sales and stock changes. Managing these activities separately can make it difficult for the owner to get a clear view of the store's daily operations and overall performance.

- Stock can be difficult to track when medicine records, batches, purchases and sales are maintained separately.
- A medicine may have multiple batches, so quantity, expiry and batch information need to remain connected.
- Billing and sales should be linked with inventory so that stock is updated automatically after a transaction.
- Supplier information needs to stay connected with purchases and the batches received from each supplier.
- Customer retail records can help the owner track purchase history, transaction frequency and total amount spent.
- Without proper sales records and analysis, it can be difficult to identify fast-moving medicines, slow-moving stock and overall sales trends.

PharmaFlowAI will address these problems by bringing the main retail operations into one system with a structured database connecting medicines, batches, suppliers, customers, sales and inventory. The system will provide a foundation for advanced sales analytics, machine learning-based insights, and a chatbot that can help the pharmacy owner access and understand store information.

4 Proposed Solution

4.1 Overview

Build a web-based pharmacy retail management system with the following components:

- **Medicine and inventory management:** maintain medicines, batches, quantities, purchase/selling prices and expiry information.
- **Billing and sales:** create bills, maintain sales records and automatically update batch-level stock after each sale.

- **Supplier management:** maintain supplier details, purchase records and the batches received from each supplier.
- **Customer retail records:** maintain basic customer information along with purchase history, transaction count and total amount spent.
- **Analytics and intelligent insights:** provide sales, inventory, supplier and customer analytics to help the owner understand store performance and make better decisions.
- **Owner dashboard and chatbot:** provide important alerts, summaries and a chatbot through which the owner can ask questions about store data.

4.2 Core Workflow

- Owner creates medicine, supplier and customer records and adds batches with quantity and expiry information.
- Stock is added through purchase/batch records and remains connected to the respective medicine and supplier.
- Owner creates a bill by selecting medicines, and the system records the sale and updates the appropriate batch inventory.
- Sales, inventory and customer transactions are stored for future analysis and reporting.
- Dashboard provides information such as low stock, expiring batches, sales performance and customer activity, while the chatbot can provide information from the stored data.

4.3 Operational constraints for an educational, small-retail setting

- Keep the system simple and usable on common computers and low-to-mid range devices through responsive web design.
- Focus on correct pharmacy retail workflows, database relationships and practical automation rather than advanced or unnecessary algorithms.
- Design the system as a modular web application so billing, analytics, ML and chatbot features can be developed and integrated gradually.

4.4 Main System Records and Responsibilities

Entity / Module	Main Records	Purpose
Owner / Admin	Login, settings, dashboard	Controls the store and views operational and analytical information.
Medicine	Name, generic name, category, manufacturer, unit, status	Maintains the master medicine catalogue.
Batch Inventory	Batch number, dates, quantity, purchase price, selling price	Tracks stock at batch level and supports expiry management.
Supplier	Name, contact, phone, email, purchases	Connects suppliers with purchases and received batches.
Customer	Name, phone, purchase history, total spent	Maintains basic retail customer transaction records.
Billing / Sales	Bill, items, quantity, price, date, total	Records retail transactions and updates inventory.
Analytics	Sales, inventory, supplier, customer metrics	Helps the owner understand store performance.
Chatbot	Natural-language queries over store data	Provides a simple way to retrieve and understand store information.

5 Solution Approach ...Engineering focus

This is an engineering course project, so the focus will be on building a reliable pharmacy retail workflow rather than developing novel machine learning algorithms. The system will connect inventory, billing, suppliers, customers and

sales records through a structured database. Advanced analytics, machine learning and chatbot features will be added on top of this foundation after the core workflow is stable.

5.1 Inventory and data quality assistance ...non-research

- Validate medicine, supplier, batch and billing records before saving them.
- Use batch-level stock tracking so quantity, purchase price and expiry information remain connected to each medicine.
- Generate alerts for low-stock, near-expiry and expired batches using configurable thresholds.
- Use FEFO-based batch selection where appropriate so batches with earlier expiry can be considered first during billing.
- Maintain stock movement history so that additions, sales, returns and manual adjustments can be traced.

5.2 Web architecture ...web-based solution

- Frontend: responsive web interface for the pharmacy owner covering dashboard, medicines, inventory, billing, suppliers, customers and analytics.
- Backend: Django application handling authentication, business logic, billing operations, inventory updates, supplier records, customer records and analytical requests.
- Database: PostgreSQL to store medicines, batches, suppliers, purchases, customers, bills, sales, stock movements and related records.

The database will act as the central source of truth so that a sale can automatically affect inventory and customer records while also becoming available for later analysis.

5.3 CI/CD and fast delivery enabler

- Automatically build the application and run tests whenever changes are pushed.
- Check database-related and backend functionality before new versions are deployed.
- Produce repeatable deployment builds for staging.
- Enable safe and frequent releases of incremental features such as inventory, billing and analytics.

6 Evaluation Criterion ...measurable and attributable

The system will be evaluated using measurable criteria connected to the main pharmacy workflow. The evaluation will focus on whether the software correctly manages retail operations, reduces manual work and provides useful information to the pharmacy owner.

6.1 Primary evaluation metric

Billing-to-inventory accuracy: percentage of completed bills for which the correct medicine quantity is deducted from the appropriate inventory batch.

- Target: $\geq 98\%$ correct inventory updates during testing.
- Attribution: measured by comparing the quantity recorded before a sale, the billed quantity and the resulting batch quantity in the database.

6.2 Secondary evaluation metrics

- Inventory accuracy: percentage of tested stock records matching the expected database quantity after purchases, sales and adjustments.
- Expiry detection: percentage of near-expiry and expired batches correctly identified by the system.
- Billing efficiency: reduction in the time required to create a bill compared with maintaining the same transaction manually.

- Analytics usefulness: ability of the system to correctly answer predefined sales, inventory, supplier and customer-related questions.
- System reliability: successful completion of core operations such as login, medicine creation, stock addition, billing and dashboard loading during the testing period.

6.3 Pilot validation plan ...high-level

- Use sample pharmacy data containing medicines, suppliers, batches, customers and realistic sales transactions.
- Run the complete workflow from purchase and stock entry to billing, inventory deduction and sales analysis.
- Compare expected results with database results for stock quantities, sales totals and customer purchase records.
- Test the system through multiple iterations and record errors or workflow issues before adding advanced features.

7 Scalability ...with foresight

The system should be designed to support growth in both the amount of pharmacy data and the number of operations performed by the owner.

1. Scalable (theoretically): The system should support increasing numbers of medicines, batches, customers, bills and stock movements by:

- using relational database relationships and properly defined foreign keys;
- adding indexes for commonly searched fields such as medicine name, batch number, expiry date and bill date;
- using pagination and filtering for large medicine, customer and sales records;
- keeping business logic separated into manageable application modules.

2. Operationally deployable (practically): The system should be deployable using common web infrastructure:

- PostgreSQL as the managed relational database;
- Django application hosted through a suitable cloud or platform-based deployment;
- containerized deployment where required;
- CI/CD pipeline for repeatable staging and production releases.

The architecture will also allow future modules such as advanced forecasting or a chatbot to be integrated without rebuilding the core inventory and billing system.

8 Engine availability heuristic

A mature set of development tools and services is available to implement PharmaFlowAI as a web-based pharmacy management system:

- Web framework: Django for backend logic, authentication, forms and web pages.
- Database: PostgreSQL for structured relationships between medicines, batches, suppliers, customers and transactions.
- Frontend: HTML, CSS and JavaScript through Django templates for the initial interface.
- Authentication: Django's built-in session-based authentication and user management.
- Analytics: Python-based data processing and visualization tools for sales and inventory analysis.
- Machine learning: established Python ML libraries can be used later for demand forecasting and related predictive tasks.
- Testing: Django testing tools and pytest where required for unit and integration testing.
- CI/CD: automated build, testing and deployment pipeline through a CI platform.

- AI chatbot: a controlled application layer that can retrieve relevant store information and provide natural-language responses to the owner.

These mature components allow the team to focus on pharmacy workflow design, database correctness, automation and measurable outcomes rather than developing infrastructure from scratch.

8.1 Technology Stack

Layer / Area	Technology	Purpose
Frontend	HTML, CSS, JavaScript, Django Templates	Responsive owner-facing web interface.
Backend	Python, Django	Application logic, authentication, forms and workflows.
Database	PostgreSQL	Structured storage and relationships across pharmacy records.
ORM / Migrations	Django ORM, Django Migrations	Database access and controlled schema changes.
Analytics	Python, Pandas, NumPy, Matplotlib	Sales, inventory, supplier and customer analysis.
Machine Learning	Scikit-learn / Python ML libraries	Later-stage forecasting and predictive analysis.
Chatbot / AI	Python AI integration layer	Natural-language access to store information.
Testing	Django Test Framework, pytest	Unit and integration testing.
Version Control	Git, GitHub	Source control and team collaboration.
CI/CD	GitHub Actions	Automated testing and deployment workflow.
Deployment	Docker + cloud/platform hosting	Repeatable deployment after the application is stable.

9 Project scope and deliverables ...iteration-friendly

9.1 Initial deliverable ...first iteration

- Medicine and supplier management with create, view, update and deactivate operations.
- Batch-level inventory with quantity, purchase price, selling price and expiry information.
- Stock movement records for additions, removals and adjustments.
- Basic inventory dashboard showing stock and expiry information.
- PostgreSQL database with properly connected medicine, supplier, batch and stock records.
- CI pipeline with application build and backend unit tests.
- Staging deployment of the working inventory module.

9.2 Subsequent deliverables

- Billing and sales workflow linked directly with batch inventory.
- Automatic stock reduction after successful billing.
- Customer retail records with purchase history, transaction count and total amount spent.
- Supplier purchase history and supplier-wise analysis.
- Improved search, filtering and pagination for medicines, customers and sales.
- Sales and inventory analytics dashboard covering revenue, fast-moving medicines, slow-moving stock and other useful indicators.
- Low-stock and expiry alerts with suggested reorder information.
- Machine learning features for demand and inventory analysis after sufficient sales data is available.

- Chatbot for natural-language queries related to inventory, billing, customers, suppliers and sales analytics.

10 Risks and mitigations

- Data accuracy: incorrect stock updates can affect billing and inventory information. Use database relationships, validation rules, controlled stock movements and automated tests.
- Incorrect batch selection: selecting the wrong batch during billing can affect expiry management and stock accuracy. Use clear batch information and FEFO-based selection where appropriate.
- User adoption: a small pharmacy owner may not want a complicated interface. Keep the main workflows simple, reduce unnecessary fields and provide clear dashboards and shortcuts.
- Customer data privacy: collect only basic retail information required for transaction history and analysis, and avoid unnecessary medical or sensitive records.
- Evaluation measurement ambiguity: define expected results for billing, stock deduction, sales totals and analytical queries before testing begins.
- Insufficient data for ML: predictive features may not be reliable without enough historical sales data. ML will therefore be treated as a later module and will not be made a dependency for the core system.
- Operational issues during deployment: use staging early and ensure CI/CD produces repeatable builds and deployments.

11 Summary

This project proposes a deployable web-based pharmacy retail management system designed to bring the main operations of a small pharmacy into one platform. The system will connect medicine and batch inventory, supplier management, billing, customer retail records and sales information through a structured PostgreSQL database.

The project meets the engineering focus by emphasizing a working business workflow, reliable database relationships, automation, testing, CI/CD and measurable evaluation rather than research-driven novelty. The initial implementation will focus on the core inventory and pharmacy workflow, followed by billing and customer records, and then advanced analytics, machine learning and chatbot functionality.

The final system will provide the pharmacy owner with a single platform for managing daily retail operations while also creating a reliable data foundation for understanding sales, inventory and customer activity. The modular design will allow advanced features to be introduced gradually without rebuilding the core system.